

Boost Converter Waveform Analysis

Set: B

Batch: G2

Group Members:

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Problem Statement

Input = 14V, Output = 20V, Switching frequency = 7.5 kHz, Output current = 1A.

Observe inductor current and switch voltage in CCM and demonstrate DCM by increasing load resistance.

Basic Definitions

CCM: Inductor current never reaches zero during a switching cycle.

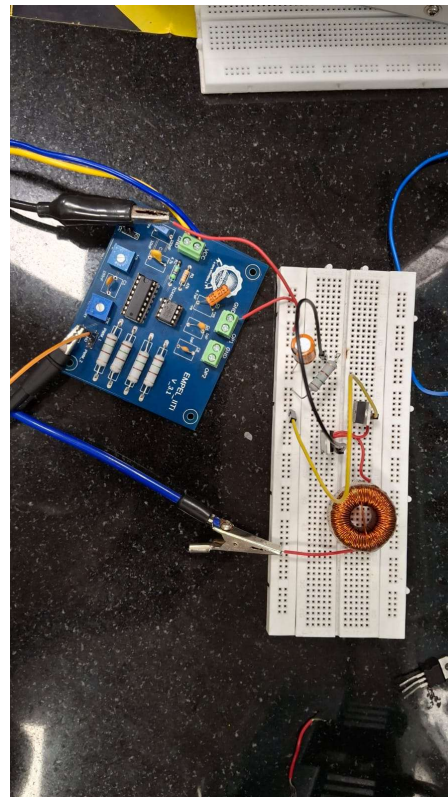
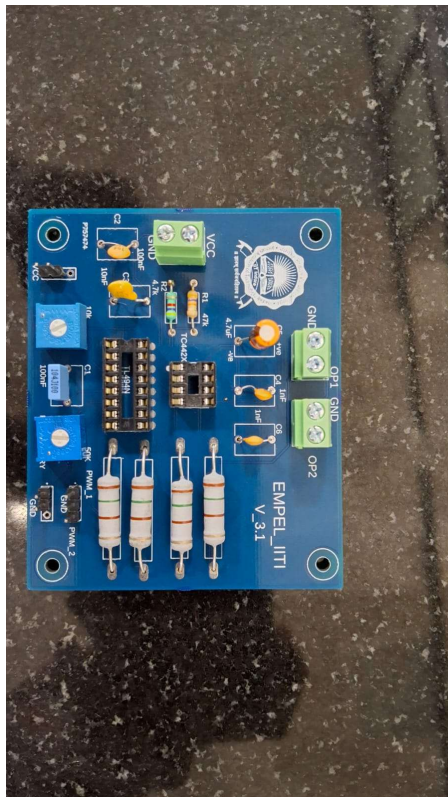
DCM: Inductor current becomes zero for part of the cycle (light load).

BCM: Boundary between CCM and DCM where current just touches zero.

Experimental Setup

PWM gate driver drives MOSFET. Boost converter consists of inductor, diode, capacitor and resistive load.

Driver Circuit and Setup



Driver Output



Measured: $f \approx 7.13 \text{ kHz}$, Duty $\approx 31.7\%$, $V_{pk} \approx 13\text{V}$

Waveforms



BCM: Inductor current just reaches zero.

CCM (20Ω): Continuous current with higher magnitude.

CCM (40Ω): Continuous but reduced current, approaching DCM.

Conclusion

Lower load resistance maintains CCM, while higher resistance shifts operation toward BCM and DCM. Observed waveforms match theoretical behavior of boost converter.